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REVIEW

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Plasmodium chitinases: revisiting a target of transmission-blockade against malaria

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Abstract

Malaria is a life-threatening disease caused by one of the five species of *Plasmodium*, among which *Plasmodium falciparum* cause the deadliest form of the disease. *Plasmodium* species are dependent on a vertebrate host and a blood-sucking insect vector to complete their life cycle. *Plasmodium* chitinases belonging to the GH18 family are secreted inside the mosquito midgut, during the ookinete stage of the parasite. Chitinases mediate the penetration of parasite through the peritrophic membrane, facilitating access to the gut epithelial layer. In this review, we describe *Plasmodium* chitinases with special emphasis on chitinases from *P. falciparum* and *P. vivax*, the representative examples of the short and long forms of this protein. In addition to the chitinase domain, chitinases belonging to the long form contain a pro-domain and chitin-binding domain. Amino acid sequence alignment of long and short form chitinase domains reveals multiple positions containing variant residues. A subset of these positions was found to be conserved or invariant within long or short forms, indicating the role of these positions in attributing form-specific activity. The reported differences in affinities to allosamidin for *P. vivax* and *P. falciparum* were predicted to be due to different residues at two amino acid positions, resulting in altered interactions with the inhibitor. Understanding the role of these amino acids in *Plasmodium* chitinases will help us elucidate the mechanism of catalysis and the mode of inhibition, which will be the key for identification of potent inhibitors or antibodies demonstrating transmission-blocking activity.

KEYWORDS

allosamidin, *Anopheles*, chitinase, malaria, *Plasmodium*, transmission-blocking

1 | INTRODUCTION

The 2020 World malaria report by the World Health organization (WHO) indicates that there were 229 million cases of malaria reported in the year 2019 with 409,000 deaths worldwide, the predominant numbers of which were from the WHO African region.[1] *P. falciparum* was responsible for 99.7% of estimated malaria cases in the WHO African region, and *P. vivax* for 75% of malaria cases in the WHO Americas.[1] In India, *P. falciparum* and *P. vivax* infections contributed to 45.6% and 53.6% of the total estimated cases in the year 2019.[1] *Plasmodium* can be transmitted to humans by the bites of infected female *Anopheles* mosquitoes, many species of which have been

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identified as vectors of disease. Preventive vector control measures like insecticide treated nets (ITNs),[2] indoor residual spraying (IRS),[3] mosquito larval source management,[4] seasonal malaria chemoprevention (SMC)[5] and intermittent preventive treatment in pregnancy (IPTp)[6] in conjunction with rapid diagnostic tests (RDT)[7] and effective medications[1,8] has significantly brought down the incidence rate for many years. The emergence of pyrethroid resistance[9] in mosquitoes and artemisinin resistance[10] in parasites has increasingly become a cause of concern for the global community. Therefore, there is an unmet need to employ alternative strategies in combination with the existing methods for malaria elimination and eradication. The potential strategies include but not restricted to drug development against novel targets,[8] vaccine development,[11,12] and transmission-blockade.[13] The aim of transmission-blocking strategy is to target *Plasmodium* proteins in the mosquito stage, consequently blocking the development of parasite in the vector.[12,13] *Plasmodium* proteins expressed at different developmental stages inside mosquitoes have been identified as targets of transmission-blockade.[12] Among the many identified targets, the parasite chitinase has an indispensable function inside the midgut of mosquitoes.[14] The secretion of chitinase by the ookinete stage of the parasite results in a cascade of events, leading to the development of oocysts. Loss of function studies validate *Plasmodium* chitinases as a promising transmission-blocking candidate.[14] A carboxy-terminal truncation of the *Plasmodium falciparum* chitinase (PfCHT1) significantly hampered oocyst development, but not gametogenesis, ex-flagellation, and ookinete formation (the stages preceding the secretion of chitinase) in *Anopheles freeborni* midguts.[14] Knockout of *Plasmodium berghei* chitinase (PbCHT1) lowered its infectivity up to 90%, due to impairment in the development of oocysts.[15] Consequently, no ookinete derived chitinase activity was detected in vitro.[15] Addition of allosamidin, a chitinase inhibitor, in the blood meal completely hampered oocyst development in *Plasmodium gallinaceum*.[16] 1C3, a murine monoclonal antibody neutralizing *P. falciparum* chitinase activity had transmission-blocking activity in mosquitoes.[17] In this report, we review *Plasmodium* chitinases with a note on the long and short forms, attempting to understand the primary basis for the differences reported between them.

2 | *PLASMODIUM* CHITINASES BELONG TO THE GH18 FAMILY OF HYDROLASES

Chitin, a long-chain polymer of N-acetyl glucosamine (NAG) is the second most abundant polysaccharide in nature.[18] Chitin can be found in three crystalline polymorphic forms–α, β, and γ, based on anti-parallel and parallel arrangements of the chitin microfibrils.[18] α-chitin, the most prevalent form of chitin, is found in fungal cell walls and the exoskeletons of insect and crustacean shells.[19] Chitin is degraded by chitinases (EC.3.2.1.14), which are glycosyl hydrolases (GH) acting on the β-1,4-linkages present between the NAG units.[20] Carbohydrate active enzyme database (CAZy) groups chitinases into GH families–18 and 19.[21] Chitinases from the GH18 family is prevalent among viruses, bacteria, fungi, plants, and animals; GH19 family among plants, bacteria (*Chitinophaga* spp.)[22] and nematodes (*Globodera rostochiensis*).[23] Studies on GH18 Chitinases have also revealed an important role for this enzyme in insect morphogenesis (*Cecropia* silk moth),[24] transmission of pathogens (*Plasmodium* spp.)[25] and vertebrate immune mechanism (Human macrophages).[26] Sequence and structural studies of *Serratia marcescens* chitinase indicated the existence of a signal peptide, serine–threonine linkers, insertion domain (α + β insertion domain) and multiple catalytic domains, which are also found in some other GH18 family of chitinases.[27,28] Further, some GH18 family chitinases also have zero to seven chitin-binding domains[29] with zero to multiple fibronectin type III domains.[30] All these elements contribute to the diversity of class 18 family of chitinases.

Chitinases belonging to GH18 families have a triosephosphate isomerase (TIM) barrel structure (Figure 1 (a),(b)). The TIM barrel fold is found in proteins with diverse functions, serving as a paradigm for one-fold with many functions scaffold structure.[32] The TIM barrel structure consists of eight beta strands (β1–β8) tethered with eight alpha helices, characteristic of a TIM barrel framework. The catalytic and substrate binding signature motifs are highly conserved within the GH18 chitinase family (Figure 2). The signature motif, DXXDXDXE, proven to have potential role of catalysis, is mapped to the β4 sheet (PDB ID: 4AXN).[34] *Plasmodium* chitinases belong to the GH18 family with a TIM barrel structure (Figure 1(a),(b)) play a key role in the transmission of the malaria parasite in the mosquito stage of its life cycle.

3 | OOKINETES EXPRESS CHITINASE TO PENETRATE THE PERITROPHIC MEMBRANE

Plasmodium parasites have a complex lifecycle comprised of sexual and asexual stages inside the vector and vertebrate host, respectively. Mosquitoes transfer the mature sporozoites to vertebrate hosts through a blood bite, which culminates with sporozoites undergoing gametogenesis. Mosquitoes ingest the male and female

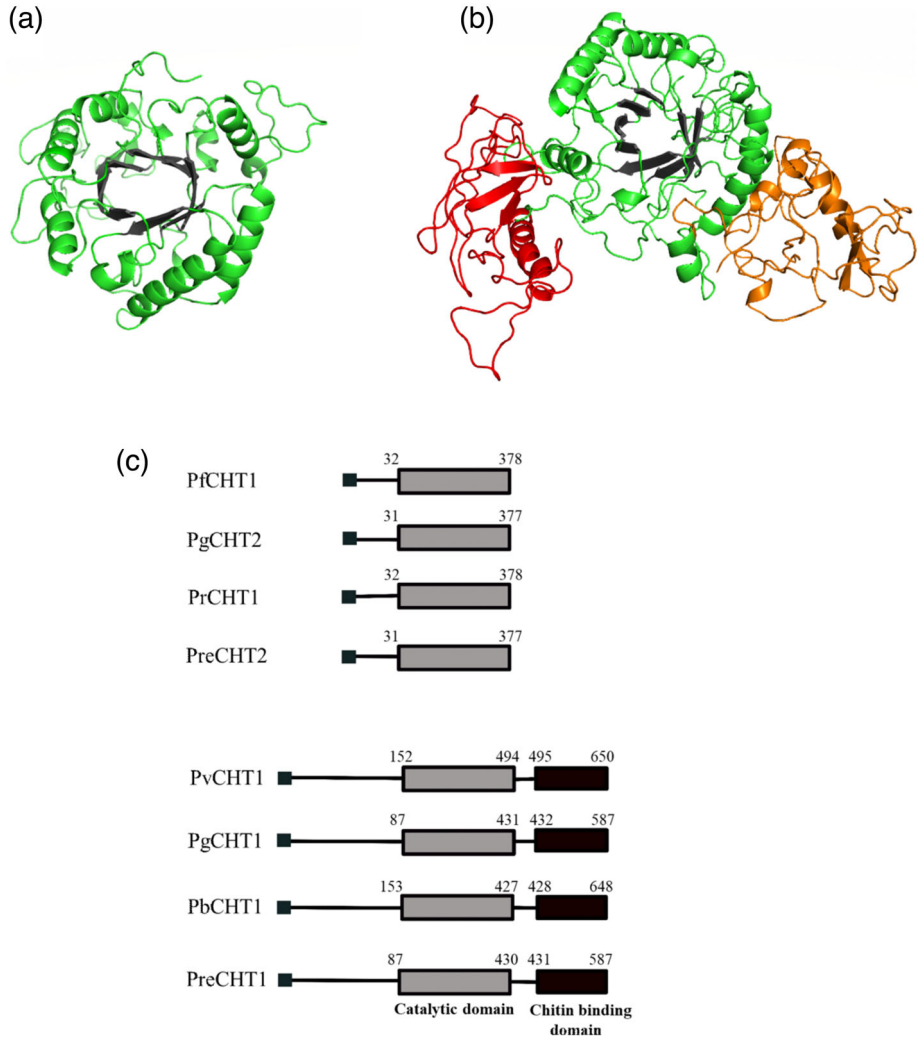

FIGURE 1 A ribbon representation of the TIM barrel fold from the predicted model structures of Chitinases from (a) *Plasmodium falciparum* and (b) *Plasmodium vivax* prepared by PyMOL. TIM Barrel is an eight stranded α-β fold with β sheets forming a barrel (black), surrounded by α helices. *Plasmodium falciparum* chitinase contains a single catalytic domain (depicted in green and black comprising of α helices and β sheets, respectively). In *Plasmodium vivax*, the pro-domain (red) and Chitin binding domain (orange) flanks the catalytic domain. (c) depicts the domain architecture of *Plasmodium* chitinases. Short form (PfCHT1, PgCHT2, PrCHT1, PreCHT2) and Long forms (PvCHT1, PgCHT1, PbCHT1, PreCHT1) of *Plasmodium* chitinases are annotated at the catalytic domain and chitin binding domains as reported elsewhere.[31] The signal sequence ranges from 1–20 (observed in the long form) to 1–31/32 residues (observed in PgCHT2, a representative of short form). Long form of chitinases harbor pro-domain, catalytic domain, and chitin binding domain. Pro-domain although not annotated in most cases, is found to be located at the N-terminal end of the catalytic domain in PvCHT1 and PbCHT1. Represented chitinases are PgCHT1 and PgCHT2: *Plasmodium gallinaceum* chitinase, PfCHT1: *Plasmodium falciparum* chitinase, PvCHT1: *Plasmodium vivax* chitinase, PbCHT1: *Plasmodium berghei* chitinase, PrCHT1: *Plasmodium reichenowei* chitinase, PreCHT1 and PreCHT2: *Plasmodium relictum* chitinase

gametocytes along with blood, which upon entering the gut, form the zygote.[35] Zygotes undergo further developmental changes to result in the formation of motile, elongated ookinetes,[36] which secrete chitinases to breakdown chitin fibrils and consequently penetrate the peritrophic membrane.[37]

Distention of the mosquito midgut post-ingestion of the blood signals the formation of a chitinous peritrophic membrane.[38] In *Anopheles stephensi*, the apical secretory vesicles of midgut epithelial cells secrete type I Peritrophic membrane (PM), which appears 12 h after blood meal and matures at the end of 48 h.[39,40] In *Anopheles gambiae*, the PM is synthesized de novo and attains maximum thickness by the end of 24 h, post- blood meal ingestion. PM is reported to be composed of chitin (NAG), although in *A. stephensi*, galactose is reported to be present along with N-acetyl galactosamine.[41] Proteins present in the PM called peritropins, contain 2,3 or

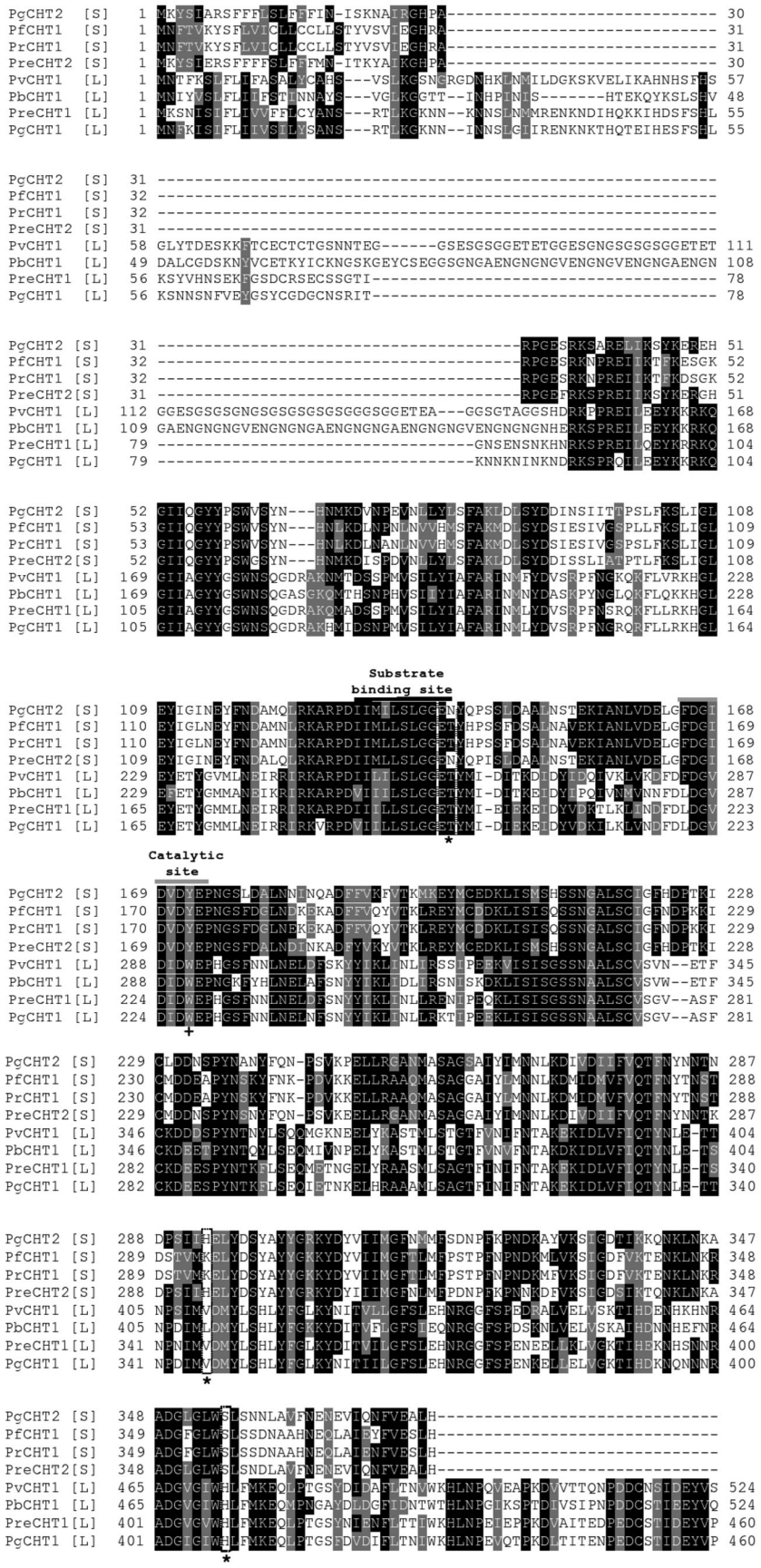

FIGURE 2 Sequence alignment of *Plasmodium* chitinases using T-Coffee and shaded with Box shade method. The long [L] and short [S] forms are indicated in parenthesis. Substrate binding site and catalytic binding sites are labeled accordingly. The residues predicted to interact with allosamidin are boxed and highlighted using asterisk (*). The catalytic site contains a position (marked as +) in which either Tyr or Trp present in short or long forms, defines the pH optimum.[33] Represented chitinases are PgCHT1 and PgCHT2: *Plasmodium gallinaceum* chitinase, PfCHT1: *Plasmodium falciparum* chitinase, PvCHT1: *Plasmodium vivax* chitinase, PbCHT1: *Plasmodium berghei* chitinase, PrCHT1: *Plasmodium reichenowei* chitinase, PreCHT1 and PreCHT2: *Plasmodium relictum* chitinases. Although, *P. knowlesi*, *P. chabaudi* and *P. yoelii* chitinases have not been included in the sequence alignment, the same trend with sequence conservation is observed at the indicated positions

4 chitin-binding domains, have been proposed to cross-link the chitin fibrils.[42] PM envelopes the food bolus and is speculated to prevent or reduce pathogen invasion (and therefore the reason why parasite secretes chitinase), facilitate digestion of blood, and protect the midgut epithelial cells from abrasion or chemical damage.[43]

The first evidence for the existence of *Plasmodium* chitinase was from an electron micrograph of *P. gallinaceum* ookinete, showcasing electron dense material at the apical end of ookinete, around which the PM appeared disrupted.[44] *P. gallinaceum* and *P. falciparum* chitinases, PgCHT1 and PfCHT1 are expressed inside the micronemes and are later accumulated at the apical end of the ookinete.[44] Expression of PgCHT1 is observed 10 h post- ex-flagellation, and by 24 h, expression is seen throughout the cytoplasm of mature ookinetes.[16] Chitinase from *P. gallinaceum* (PgCHT1) is produced as a partially active or inactive zymogen, which later gets activated by the proteases in the mosquito midgut.[45] *P. vivax* and *P. berghei* ookinetes secrete chitinase as a zymogen and therefore are like PgCHT1.[15,46] On the other hand, *P. falciparum* ookinetes secrete chitinase (PfCHT1) as an enzymatically active form. The localisation of trypsin within the PM validates the dependency of PgCHT1 on proteases in *P. gallinaceum*.[25,47] Addition of a serine protease inhibitor,[16] leupeptin or anti-trypsin antisera was found to block the activation of PfCHT1 and transmission of the parasite.[47,48] It is speculated that PfCHT1 is an ortholog of the *P. gallinaceum* short form chitinase, PgCHT2.[31] After the chitin layer is breached, the ookinetes reach the basal lamina and develop to form the oocysts harboring sporozoites. Upon rupture of oocysts, the sporozoites travel to the salivary gland and enter the vertebrate host following a mosquito bite. Therefore, dissolution of the peritrophic membrane is an essential step facilitated by chitinases secreted by *Plasmodium*.[49]

4 | PROPERTIES OF *PLASMODIUM* CHITINASES

Chitinases from *Plasmodium* species have been reported to exist in two different forms, long form, and a short form. Phylogenetic analysis predicts *P. gallinaceum* chitinases to be the ancestor of *Plasmodium* chitinases present in species specifically infecting mammals. While both forms of chitinases have been identified in *P. gallinaceum*, either short form or the long form has been annotated for *P. falciparum* and *P. vivax*, respectively. *P. vivax* chitinase contains pro-domain, chitin-binding and catalytic domains, whereas *P. falciparum* chitinase contains only the catalytic domain (Figure 1 (c)). Further, studies with other species of *Plasmodium* (*P. knowlesi, P. chabaudi, P. yoelii, P. reichenowi*) indicated that possibly these *Plasmodium* spp. subsequently lost the short or long forms, the evolutionary advantage of which is still unclear.[31,50] Characterisation of PvCHT1, PgCHT1, PreCHT1 (*P. relictum* chitinase) and PbCHT1 (*P. berghei* chitinase) indicated that long forms of chitinases are expressed as catalytically inactive forms,[16] which subsequently gets activated within the mosquito midgut by proteases.[50,51] In fact, a dual lysine repeat (KK) present at the junction of the catalytic domain, 42 amino acids from the N-terminal end of PgCHT1 (and corresponding position in PvCHT1, PgCHT1, and PbCHT1) can be a plausible cleavage site for trypsin. In *A. gambiae* chitinase, a 13-mer peptide derived from a putative pro-domain region was demonstrated to inhibit both *A. gambiae* and *P. falciparum* chitinases.[52] Chitin-binding domains are carbohydrate binding modules, typically found in multi-domain proteins along with other domains. There are three types of Carbohydrate-binding modules (CBM)–Type A, B, and C. Type A binds to chitin and other polysaccharides, which is further classified into CBM1, CBM2, CBM3, CBM5, and CBM10.[53] Although, CBM2, CBM3, and CBM5 were shown to have chitin-binding properties, only CBM5 is found in chitinases. Mutational analysis on chitin binding domains of *Bacillus circulans* and *Streptomyces griseus* chitinases highlighted the importance of this domain in substrate specificity and catalysis.[54,55]

Experimental analysis using varying lengths of the substrate, NAG, showed that recombinant *P. falciparum* chitinase (rPfCHT1) efficiently processes polymers containing 3, 4, or 5 units of NAG.[46] Likewise, rPvCHT1 was also demonstrated to hydrolyse chitotrioside containing 4-methylumbelliferone (4-MU-GlcNAc$_3$),[46] indicating that PfCHT1 and PvCHT1 are endochitinases, like PgCHT1.[46] Maximal activity was observed at pH 4.5 and pH 7.0– 8.0 for rPfCHT1 and rPvCHT1, respectively.[46] Site-directed mutagenesis of the catalytic residue, Trp291, present in between Asp290 and Glu292 at the active site (numbering with respect to PvCHT1), to Tyr in *Manduca sexta* chitinase (a long form chitinase belonging to the GH18 family) resulted in a decrease in the pH optimum by one unit.[33] Plant chitinases derived from *Hevea brasiliensis* and *Cycas revoluta* have a pH optimum of 4.0 – 5.0 and Tyr residue at this position.[56,57] Sequence analysis of the long and short forms of chitinases clearly showed that Trp291 is conserved in the long form, whereas Tyr291 (Residue numbering of PvCHT1) was present in the short form of chitinases (Figure 2), inferring the importance of this amino acid in conferring optimum pH. Whether this observation can be extrapolated to other chitinases needs to be ascertained. Knowledge

about amino acids that can change the pH optima of chitinases has potential industrial applications. Optimal temperature for activity was found to be 40°C and 35°C for rPfCHT1 and rPvCHT1, respectively, at pH 8.0. The specific activities of purified rPfCHT1 and rPvCHT1 were 2070 x 10^3 and 7313 units, respectively,[46] where each unit for specific activity of recombinant chitinases was defined as fluorescence units per min per mg of protein using 4MU-$GlcNAc_3$ as a substrate. There appears to be a correlation between the long form of chitinases containing the chitin-binding domain and low specific activities, exemplified by the *P. vivax* chitinase. Whether this correlation is observed in other chitinases needs to be determined. It will be interesting to understand the reason as to why the long forms are secreted as inactive zymogens and the short forms are secreted as catalytically active forms. Possibly the presence of chitin-binding domains along with the secretion of inactive forms, introduces a spatio-temporal regulation for the long forms of chitinases. This regulation is dialed out in short forms along probably compensated with high specific activity. A recent report showed that PgCHT2 and its ortholog PfCHT1 form high-molecular weight (HMW) complexes, the implications of which are unknown.[58]

Sequence alignment of four representative long forms (PvCHT1, PbCHT1, PreCHT1, and PgCHT1) or four short forms (PgCHT2, PfCHT1, PrCHT1, and PreCHT2), lead to the construction of a consensus-like signature sequence for the long and short forms. When the consensus sequence belonging to the long and short forms were aligned, some of the positions were not invariant (Figure 3). For example, the position 283, immediately preceding the catalytic site, contains Gly (nonpolar residue) in the short form and Asp (negatively charged residue) in the long form (Figure 2). Although, the reason behind form-specific conservation of amino acid residues at certain positions are not clearly understood, its plausible role in defining the properties of long or short forms cannot be undermined. Amino acid residue differences at some of these positions highlighted in Figure 3, could possibly lead to an alteration in the specific activity of chitinases. Parallel to this observation, pH optimum can be partly attributed to a Trp or Tyr at position 291 (labeled as + in Figure 2).

5 | INHIBITION OF *PLASMODIUM* CHITINASES WITH ALLOSAMIDIN

Allosamidin, a pseudo-trisaccharide isolated from the mycelium of *Streptomyces* species is a potent inhibitor of chitinases.[59] The prevalence of oocysts was significantly decreased when allosamidin was fed along with the blood meal containing *P. gallinaceum* and *P. falciparum* gametocytes to *Aedes aegypti* and *Anopheles freeborni*, respectively.[45] The addition of exogenous chitinase or polyoxin D, an inhibitor of chitin synthase, to the blood meal alleviated the allosamidin-mediated inhibition of oocyst development,[60] validating the target of allosamidin as chitinase. The inhibitory concentration (IC_{50}) for allosamidin against PfCHT1 and PgCHT2 (short forms) was in the range of 1 – 300 nM,[46] whereas against PvCHT1 and PgCHT1 (long forms) was in the range of 6 – 12 μM.[46] In the absence of structural information, the reason behind this differential inhibition of allosamidin of *P. vivax* and *P. falciparum/gallinaceum* chitinases is not clear.

Homology modeling of chitinases from *P. vivax* and *P. falciparum* using Phyre[261] generated a structure containing the expected $(\alpha/\beta)_8$ TIM barrel backbone with the catalytic and substrate-binding sites mapped onto β4 and β3 sheets, respectively. Predictive structural modeling studies of PfCHT1 and PvCHT1 with allosamidin performed using AutoDock 4.2[62] showed that Glu259 and Thr260 (numbering with reference to PvCHT1) are involved in polar hydrogen bonding interactions with –

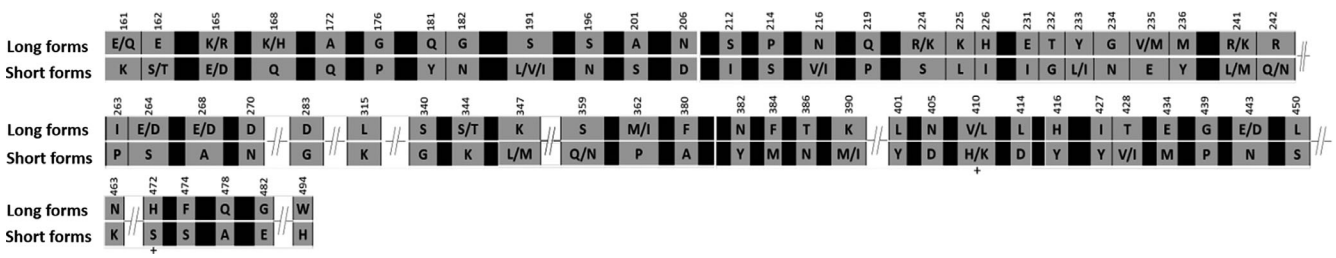

FIGURE 3 Comparison of the consensus amino acid sequences of the long and short forms of *Plasmodium* chitinases. The individual consensus sequence for long and short forms were derived by sequence alignment of chitinases from four species each (the sequences of which are compared in Figure 2). At the positions displayed in the figure, differences in amino acids were observed. For example, position 283 contains Gly (nonpolar residue) in the short and Asp (negatively charged residue) in the long form. + indicates the predicted residue position involved in binding to allosamidin. Residue numbering is with respect to PvCHT1

OH of tetrahydro-4H-cyclopenta[d]oxazole and –CH_2OH of the middle glucose ring of allosamidin, respectively (Figure 4). Lys410 in PfCHT1 is involved in a hydrogen bonding interaction with -OH of the terminal glucose ring of allosamidin. This interaction is not possible in PvCHT1 with a valine residue at the same position. In PfCHT1, an amide moiety substituted on the middle glucose ring of allosamidin is likely to fit optimally in the binding pocket due to the presence of the comparatively smaller and less bulky residue, Ser472. Owing to the presence of a slightly bulkier residue like His at this position in PvCHT1, the substituted amide on the middle glucose ring of allosamidin might not fit optimally in PvCHT1. Therefore, out of the four residues predicted to bind to allosamidin, residues at two positions (410 and 472) were not conserved between PfCHT1 and PvCHT1, possibly the reason behind differential sensitivity to allosamidin (Figures 2 and 3). When the other sequences of short and long forms were analyzed, the following trend was observed: Lys or His, and Val or Leu were present at 410th position in the short and long forms, respectively; Ser and His were present at 472nd position in the short and long forms, respectively (displayed as * and boxed in Figure 2). Whether allosamidin exhibits low and high binding affinities to all the long and short forms of chitinases from different *Plasmodium* spp., needs to be experimentally determined.

6 | SUMMARY

Transmission-blocking strategies aim to inhibit parasite development inside the mosquito host by targeting parasite proteins. Inside the midgut of the mosquitos, the parasite secretes chitinases belonging to the GH18 hydrolase family to degrade the peritrophic membrane and invade the epithelial layer. Different species of *Plasmodium* secrete chitinases either as a catalytically active form as in *Plasmodium falciparum* (short form) and/or as a partially active or inactive zymogen, as in *Plasmodium vivax* (long form). *P. gallinaceum* has both forms with the evolutionary descendants retaining either one form or the other. *P. falciparum* chitinase exhibits ~283-fold and ~1000-fold differences in specific activity and sensitivity to allosamidin, respectively, as compared to its *P. vivax* counterpart. The observed differences could be due to changes in the amino acid residues mapped at specific positions not restricted to the catalytic and substrate-binding sites (Figure 3). *Plasmodium* chitinases can serve as a paradigm to study the sequence–structure–function correlation of proteins and therefore it will be interesting to perform mutational and structural studies to understand the role of amino acid positions exhibiting form-specific conservation in the long and short forms. Further, this understanding can facilitate the design of inhibitors or selection of epitopes for antibodies with transmission-blocking activity.

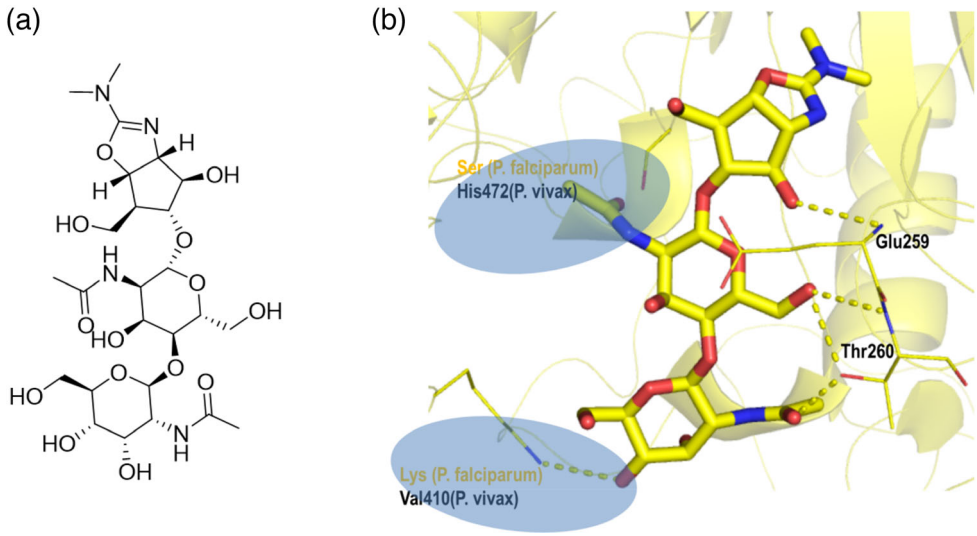

FIGURE 4 Prediction on binding mode of allosamidin with PfCHT1. (a) Two-dimensional structural representation of allosamidin ($C_{25}H_{42}N_4O_{14}$). (b) Representation of binding mode of allosamidin to *Plasmodium falciparum* chitinase showing conserved residues – Glu259 and Thr260 forming polar hydrogen bonding interactions with –OH of tetrahydro-4H-cyclopenta[d]oxazole. Unlike His472 in PvCHT1, Ser472 in PfCHT1 orients the amide moiety of allosamidin for optimal interaction. Lys410 (Pf) interacts with the terminal glucose ring in allosamidin whereas His472 and Val410 of *Plasmodium vivax* (Pv) chitinase fails to make polar interactions. Residue numbering is with respect to PvCHT1

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CONFLICT OF INTEREST

All authors have no conflicts of interest to declare relevant to this study.

AUTHOR CONTRIBUTIONS

Vysakh K. Viswanath: Data curation; writing-original draft; writing-review & editing. **Suraj T. Gore:** Investigation. **Ashwathi Valiyaparambil:** Data curation; writing-original draft. **Subhendu Mukherjee:** Investigation. **Anirudha Lakshminarasimhan:** Supervision; writing-review & editing.

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